MSHA Accident and Injury Database

Structured-Data EDA and Narrative NLP Report

Generated September 23, 2026 at 14:16

```
Records ..................... 274,955
Columns ..................... 57
Date coverage ............... 2000-01-01 to 2026-09-15
Memory footprint ............ 806.3 MB
Exact duplicate rows ......... 0

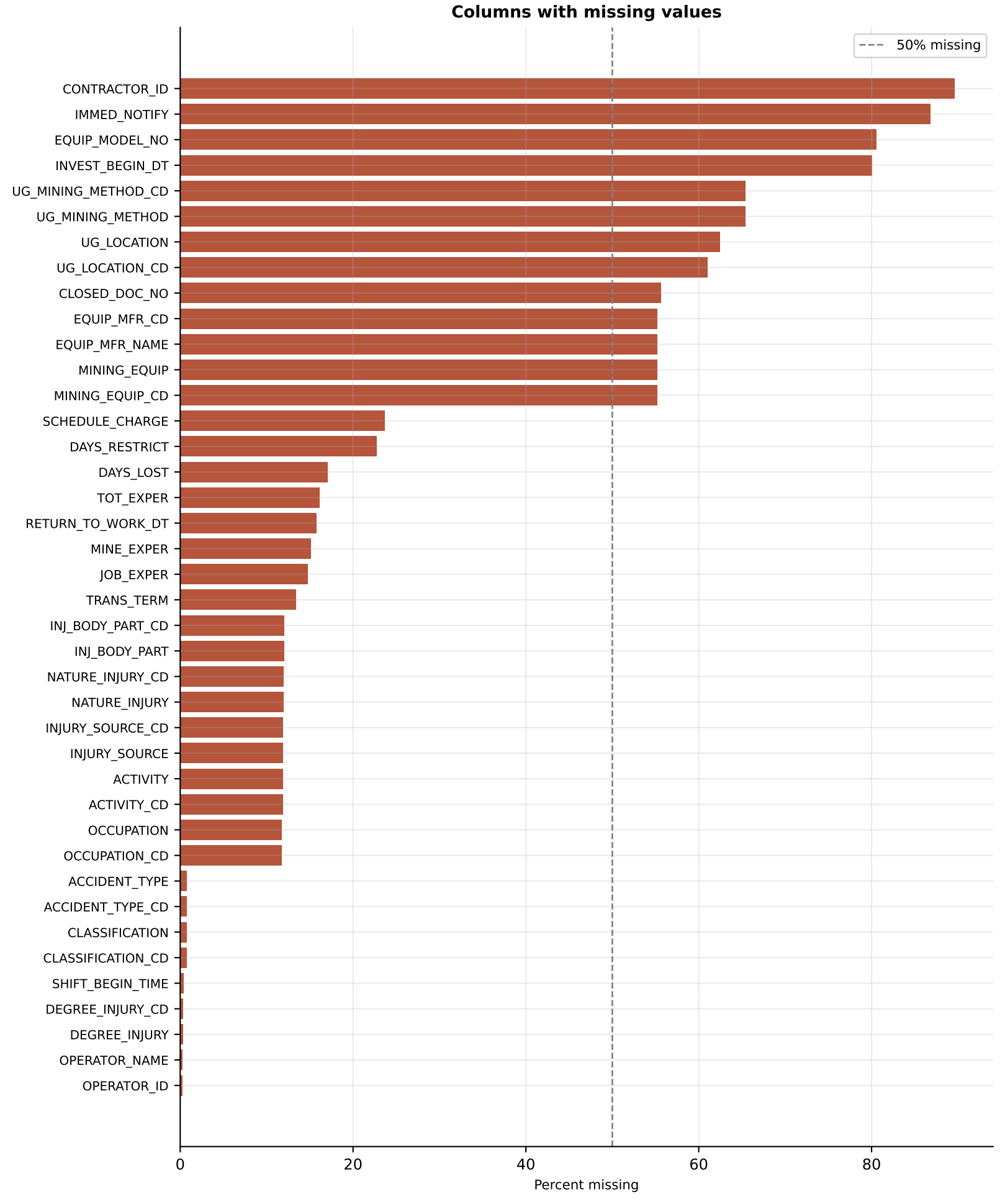
Identifier fields ........... 7
Code fields ................. 15
Descriptive label fields ..... 18
Numeric fields .............. 13
Date fields ................. 3

Report sections
  1. Schema and completeness
  2. Missingness
  3. Numeric distributions and correlation
  4. Categorical distributions
  5. Code-to-label integrity
  6. Temporal patterns
  7. Narrative descriptive EDA
  8. N-grams and word cloud
  9. TF-IDF and NMF topics
 10. Narrative/outcome comparisons
 11. Findings and modeling recommendations
```

1. Schema and Data-Quality Profile

column	dtype	non_null	missing	missing_%	unique	example
MINE_ID	string	274955	0	0.00	13533	0100003
CONTROLLER_ID	string	274234	721	0.26	6670	0041044
CONTROLLER_NAME	object	274234	721	0.26	6676	Lhoist Group
OPERATOR_ID	string	274234	721	0.26	8027	L13586
OPERATOR_NAME	object	274234	721	0.26	10222	Lhoist North America of Alabama, LL
CONTRACTOR_ID	string	28656	246299	89.58	5177	1KJ
DOCUMENT_NO	string	274955	0	0.00	274955	220152230015
SUBUNIT_CD	string	274955	0	0.00	10	03
SUBUNIT	object	274955	0	0.00	10	STRIP, QUARY, OPEN PIT
ACCIDENT_DT	datetime64[ns]	274955	0	0.00	9744	2015-07-27 00:00:00
CAL_YR	int64	274955	0	0.00	27	2015
CAL_QTR	int64	274955	0	0.00	4	3
FISCAL_YR	int64	274955	0	0.00	27	2015
FISCAL_QTR	int64	274955	0	0.00	4	4
ACCIDENT_TIME	float64	274954	1	0.00	1443	740.0
DEGREE_INJURY_CD	string	274045	910	0.33	11	06
DEGREE_INJURY	object	274045	910	0.33	11	NO DYS AWY FRM WRK,NO RSTR ACT
FIPS_STATE_CD	string	274955	0	0.00	54	01
UG_LOCATION_CD	string	107228	167727	61.00	8	04
UG_LOCATION	object	103271	171684	62.44	7	INTERSECTION
UG_MINING_METHOD_CD	string	95193	179762	65.38	7	05
UG_MINING_METHOD	object	95193	179762	65.38	7	Continuous Mining
MINING_EQUIP_CD	string	123212	151743	55.19	69	52
MINING_EQUIP	object	123212	151743	55.19	69	Road grader, Motor grader, Motor pa
EQUIP_MFR_CD	string	123159	151796	55.21	245	018
EQUIP_MFR_NAME	object	123159	151796	55.21	193	Caterpillar
EQUIP_MODEL_NO	string	53371	221584	80.59	21068	H140
SHIFT_BEGIN_TIME	float64	273965	990	0.36	697	445.0
CLASSIFICATION_CD	string	272849	2106	0.77	28	17
CLASSIFICATION	object	272849	2106	0.77	28	MACHINERY
ACCIDENT_TYPE_CD	string	272845	2110	0.77	44	02
ACCIDENT_TYPE	object	272845	2110	0.77	44	Struck against a moving object
NO_INJURIES	int64	274955	0	0.00	16	1
TOT_EXPER	float64	230599	44356	16.13	3348	5.38
MINE_EXPER	float64	233477	41478	15.09	3229	5.38
JOB_EXPER	float64	234309	40646	14.78	3047	12.5
OCCUPATION_CD	string	242571	32384	11.78	255	375
OCCUPATION	object	242571	32384	11.78	203	Grader operator, Roadgrader operato
ACTIVITY_CD	string	242259	32696	11.89	99	054
ACTIVITY	object	242259	32696	11.89	99	Grader
INJURY_SOURCE_CD	string	242201	32754	11.91	127	076
INJURY_SOURCE	object	242201	32754	11.91	127	SURFACE MINING MACHINES
NATURE_INJURY_CD	string	241974	32981	12.00	38	330
NATURE_INJURY	object	241974	32981	12.00	38	SPRAIN,STRAIN RUPT DISC
INJ_BODY_PART_CD	string	241973	32982	12.00	46	450
INJ_BODY_PART	object	241973	32982	12.00	46	SHOULDERS (COLLARBONE/CLAVICLE/SCAP
SCHEDULE_CHARGE	float64	209934	65021	23.65	227	0.0
DAYS_RESTRICT	float64	212526	62429	22.71	448	0.0
DAYS_LOST	float64	227988	46967	17.08	666	0.0
TRANS_TERM	string	238180	36775	13.37	2	N
RETURN_TO_WORK_DT	datetime64[ns]	231615	43340	15.76	9691	2015-07-28 00:00:00
IMMED_NOTIFY_CD	string	274955	0	0.00	14	?
IMMED_NOTIFY	object	36274	238681	86.81	12	SERIOUS INJURY
INVEST_BEGIN_DT	datetime64[ns]	54838	220117	80.06	8728	2003-10-22 00:00:00
NARRATIVE	object	274943	12	0.00	273395	Employee was clearing rocks from th
CLOSED_DOC_NO	string	122033	152922	55.62	122016	320102430005
COAL_METAL_IND	string	274955	0	0.00	2	M

2a. Missing Values by Column



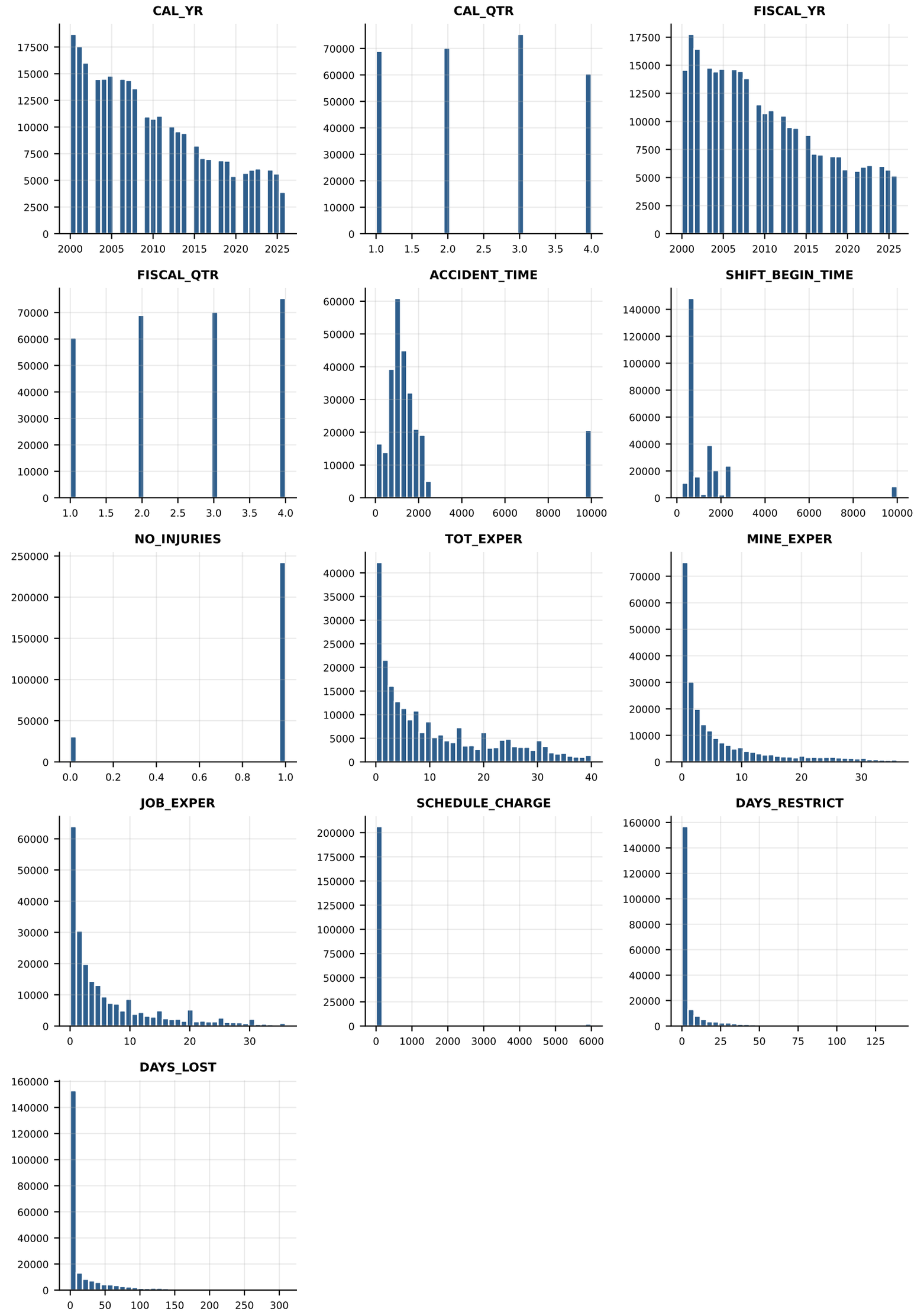
2b. Missing-Value Summary

	missing_count	missing_%
CONTRACTOR_ID	246299	89.58
IMMED_NOTIFY	238681	86.81
EQUIP_MODEL_NO	221584	80.59
INVEST_BEGIN_DT	220117	80.06
UG_MINING_METHOD_CD	179762	65.38
UG_MINING_METHOD	179762	65.38
UG_LOCATION	171684	62.44
UG_LOCATION_CD	167727	61.00
CLOSED_DOC_NO	152922	55.62
EQUIP_MFR_CD	151796	55.21
EQUIP_MFR_NAME	151796	55.21
MINING_EQUIP	151743	55.19
MINING_EQUIP_CD	151743	55.19
SCHEDULE_CHARGE	65021	23.65
DAYS_RESTRICT	62429	22.71
DAYS_LOST	46967	17.08
TOT_EXPER	44356	16.13
RETURN_TO_WORK_DT	43340	15.76
MINE_EXPER	41478	15.09
JOB_EXPER	40646	14.78
TRANS_TERM	36775	13.37
INJ_BODY_PART_CD	32982	12.00
INJ_BODY_PART	32982	12.00
NATURE_INJURY_CD	32981	12.00
NATURE_INJURY	32981	12.00
INJURY_SOURCE_CD	32754	11.91
INJURY_SOURCE	32754	11.91
ACTIVITY	32696	11.89
ACTIVITY_CD	32696	11.89
OCCUPATION	32384	11.78
OCCUPATION_CD	32384	11.78
ACCIDENT_TYPE	2110	0.77
ACCIDENT_TYPE_CD	2110	0.77
CLASSIFICATION	2106	0.77
CLASSIFICATION_CD	2106	0.77
SHIFT_BEGIN_TIME	990	0.36
DEGREE_INJURY_CD	910	0.33
DEGREE_INJURY	910	0.33
OPERATOR_NAME	721	0.26
OPERATOR_ID	721	0.26
CONTROLLER_ID	721	0.26
CONTROLLER_NAME	721	0.26
NARRATIVE	12	0.00
ACCIDENT_TIME	1	0.00

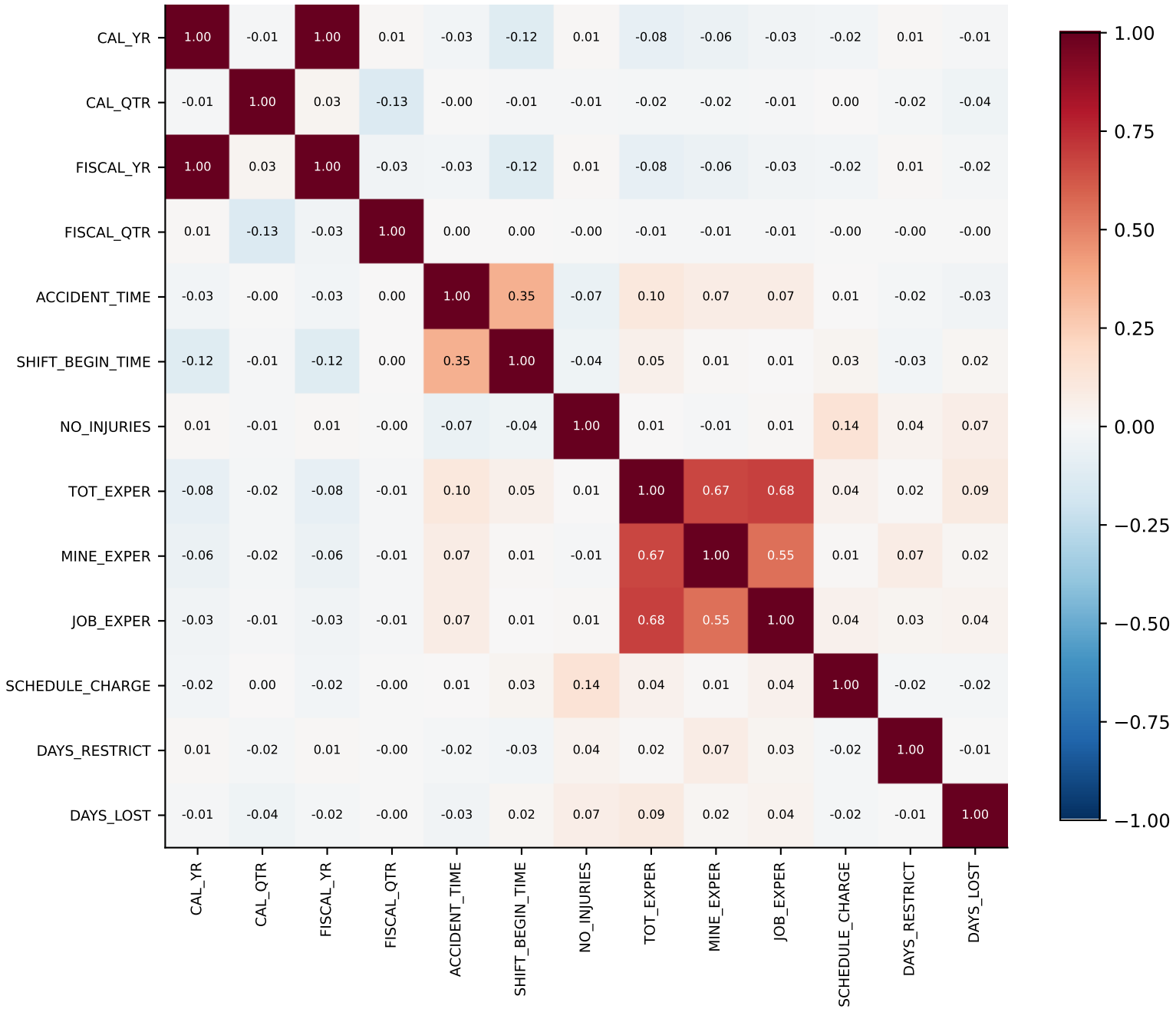
3a. Numeric Variable Summary

	count	mean	std	min	25%	50%	75%	max	missing	missing_%	zero_%	skew
CAL_YR	274955.0	2009.89	7.38	2000.00	2004.00	2008.00	2015.00	2026.0	0	0.00	0.00	0.51
CAL_QTR	274955.0	2.46	1.09	1.00	1.00	2.00	3.00	4.0	0	0.00	0.00	0.02
FISCAL_YR	274955.0	2010.11	7.39	2000.00	2004.00	2009.00	2015.00	2026.0	0	0.00	0.00	0.50
FISCAL_QTR	274955.0	2.58	1.11	1.00	2.00	3.00	4.00	4.0	0	0.00	0.00	-0.09
ACCIDENT_TIME	274954.0	1855.99	2381.77	0.00	843.00	1215.00	1720.00	9999.0	1	0.00	0.01	2.94
SHIFT_BEGIN_TIME	273965.0	1307.38	1654.17	0.00	630.00	700.00	1530.00	9999.0	990	0.36	0.05	4.45
NO_INJURIES	274955.0	0.91	0.58	0.00	1.00	1.00	1.00	36.0	0	0.00	11.13	31.45
TOT_EXPER	230599.0	10.94	10.89	0.01	2.00	7.00	18.00	66.0	44356	16.13	0.00	1.01
MINE_EXPER	233477.0	6.45	8.62	0.01	0.69	2.76	8.44	65.0	41478	15.09	0.00	1.88
JOB_EXPER	234309.0	6.88	8.47	0.01	1.00	3.17	10.00	66.0	40646	14.78	0.00	1.76
SCHEDULE_CHARGE	209934.0	69.07	620.41	0.00	0.00	0.00	0.00	6000.0	65021	23.65	74.40	9.32
DAYS_RESTRICT	212526.0	9.09	30.85	0.00	0.00	0.00	4.00	3310.0	62429	22.71	53.24	17.32
DAYS_LOST	227988.0	27.65	64.93	0.00	0.00	0.00	22.00	3616.0	46967	17.08	42.16	5.74

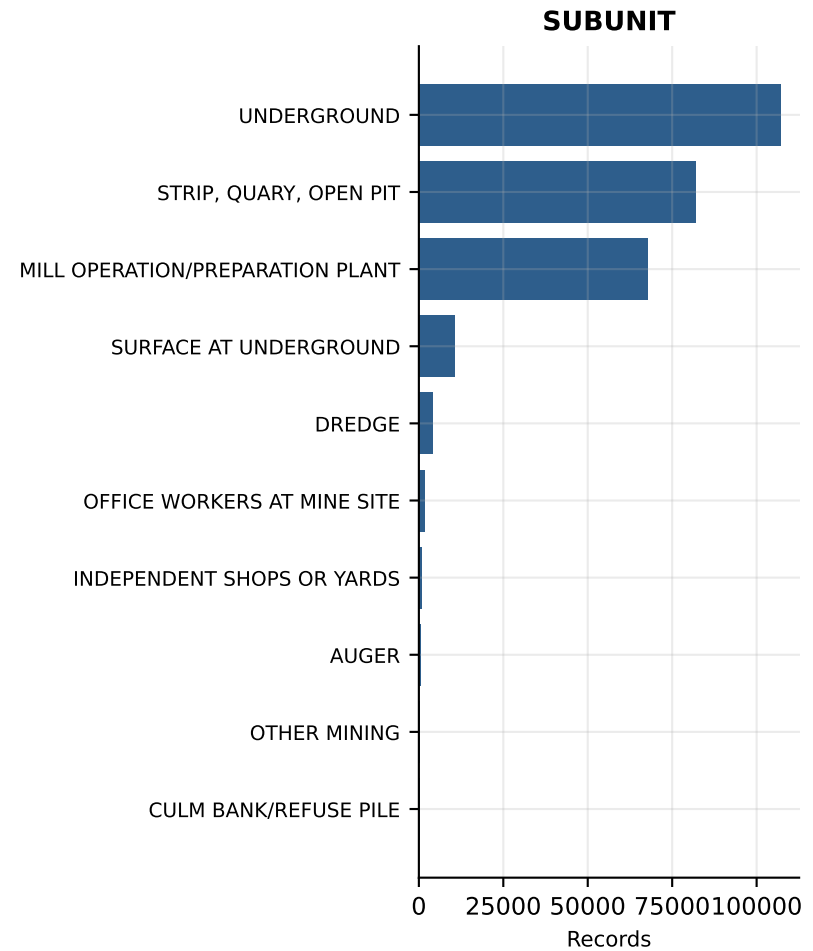
3b. Numeric Variable Distributions



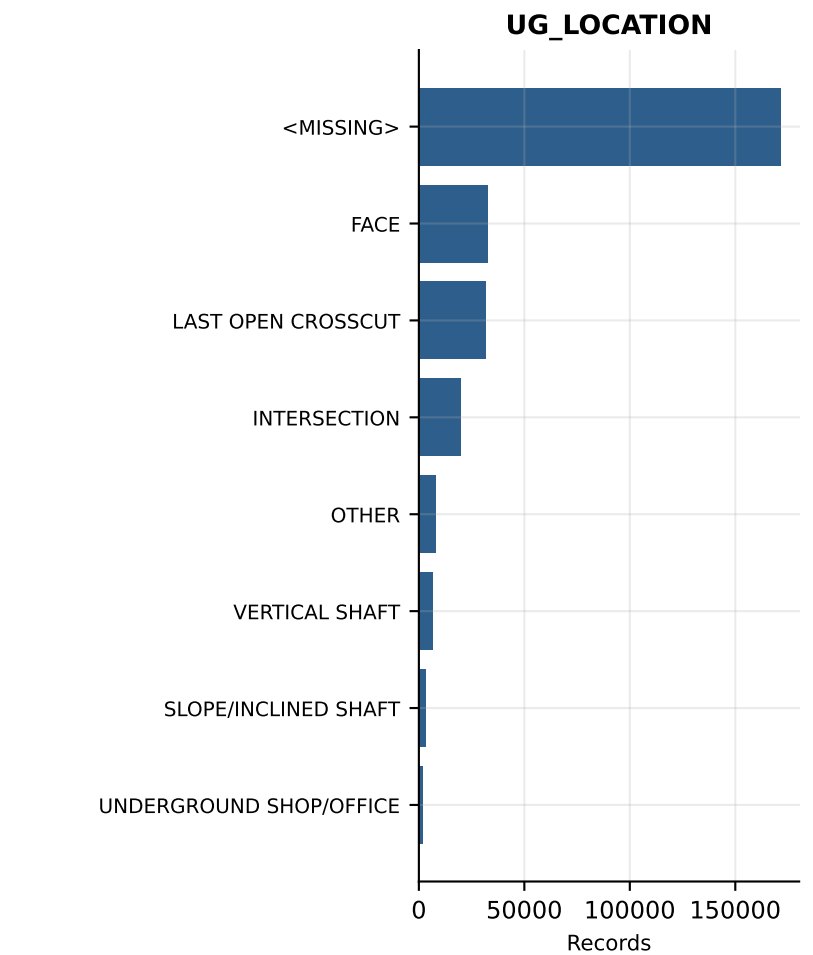
3c. Numeric Correlation Matrix

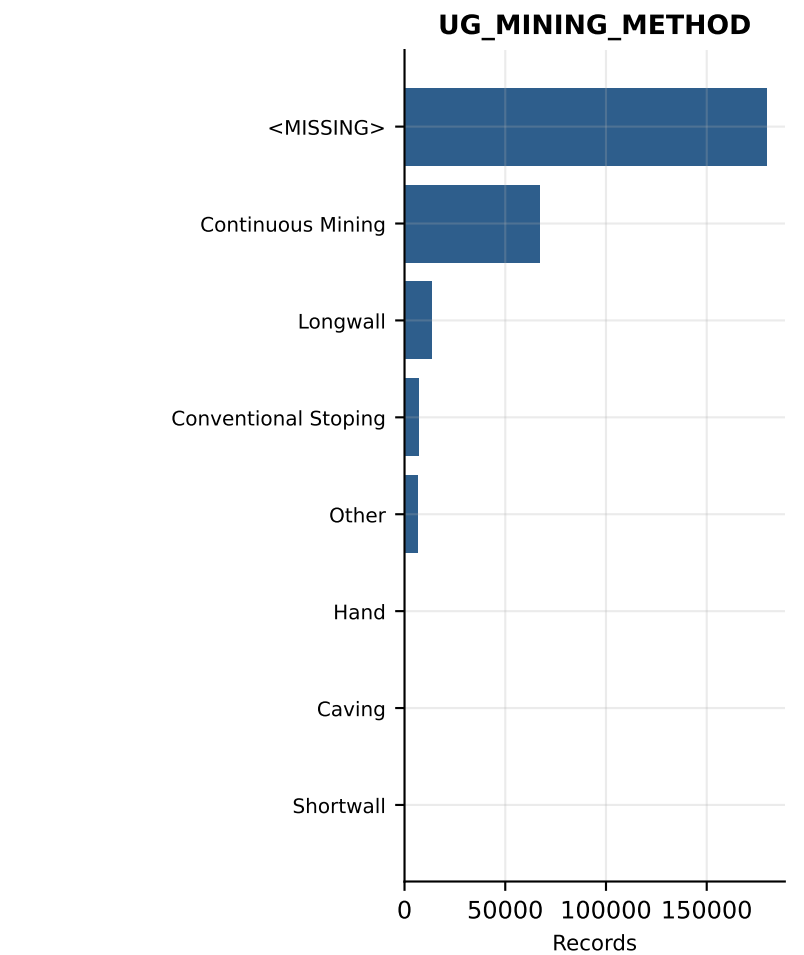


4. Categorical Variable Distributions

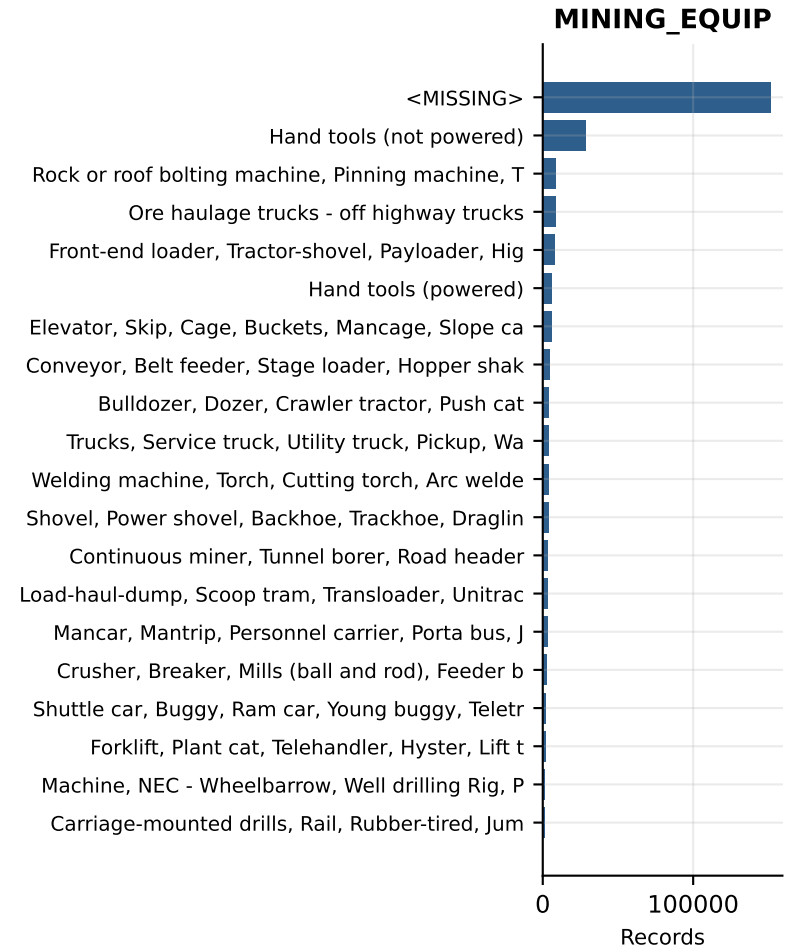


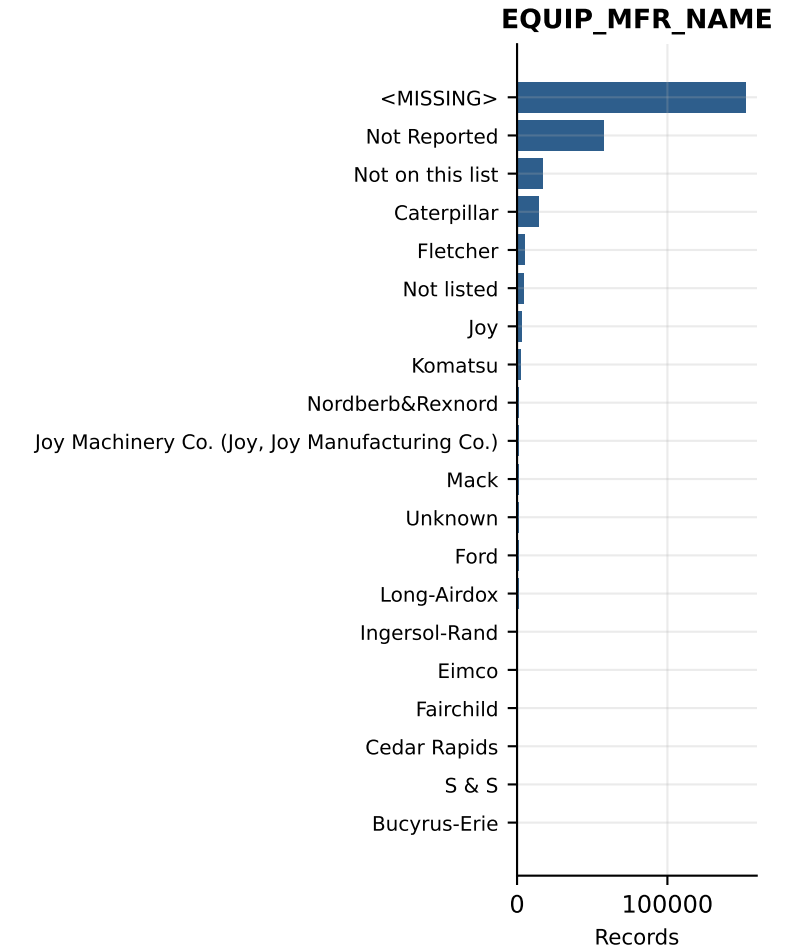


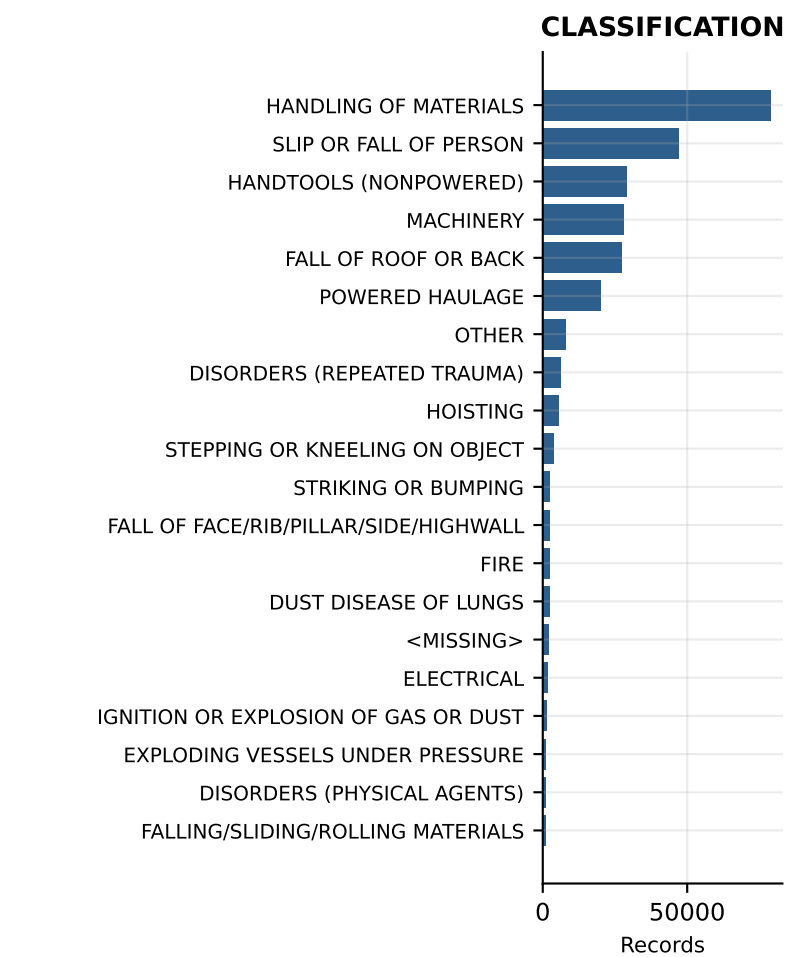


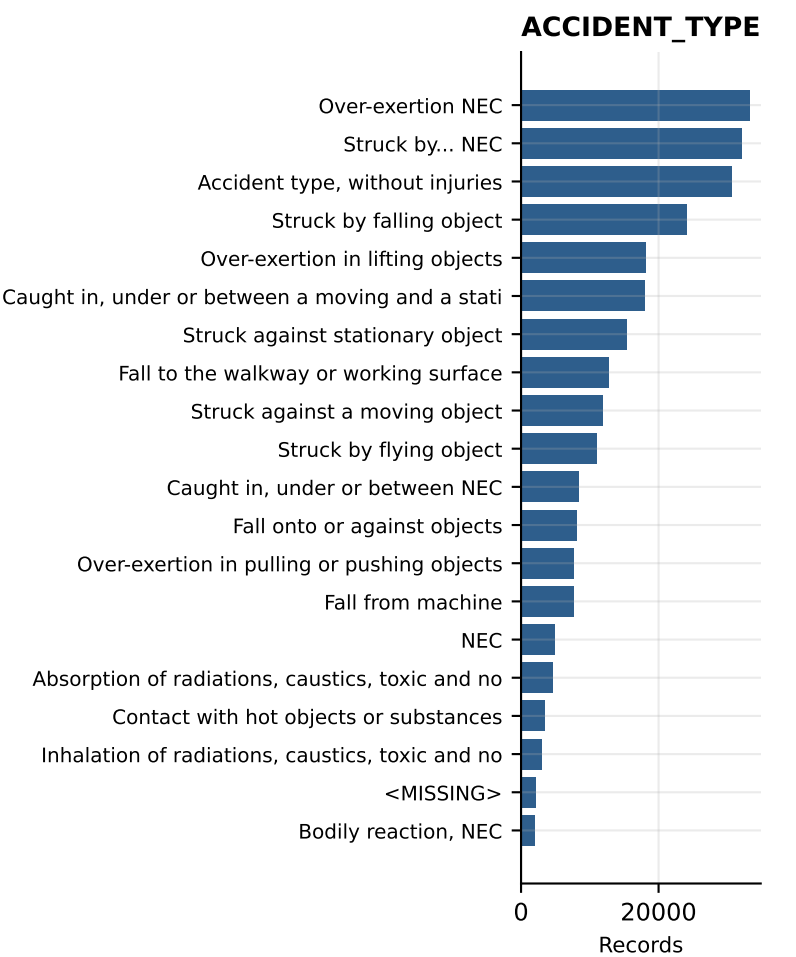


4. Categorical Variable Distributions

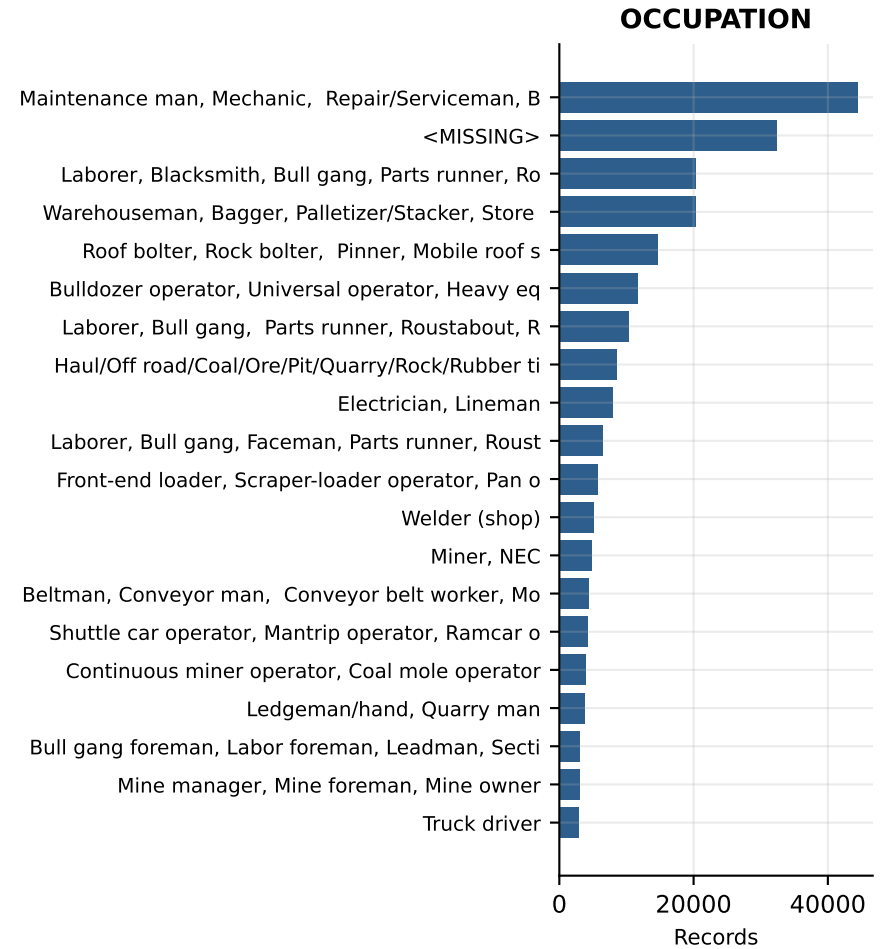


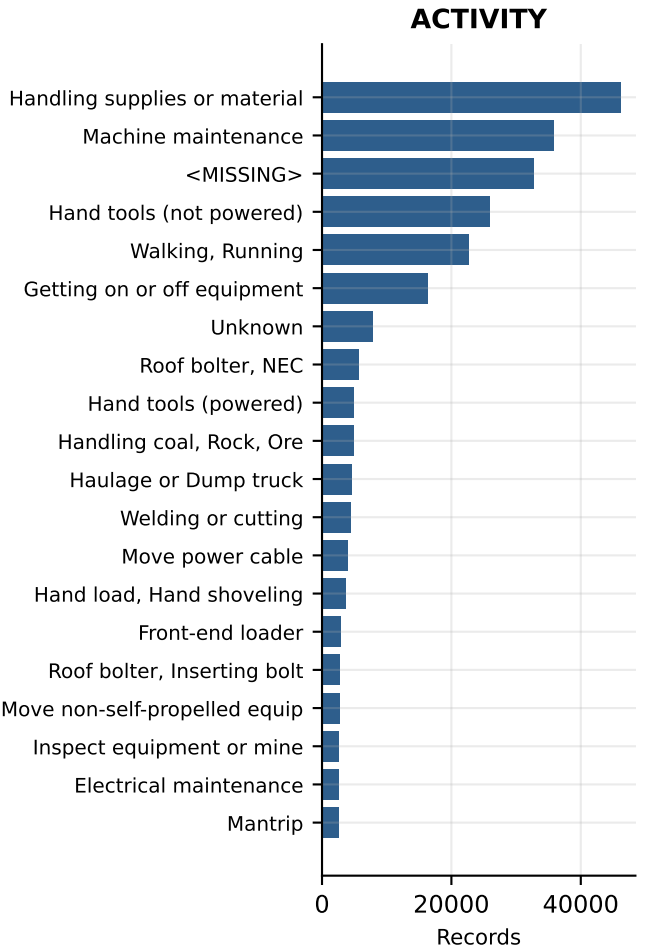


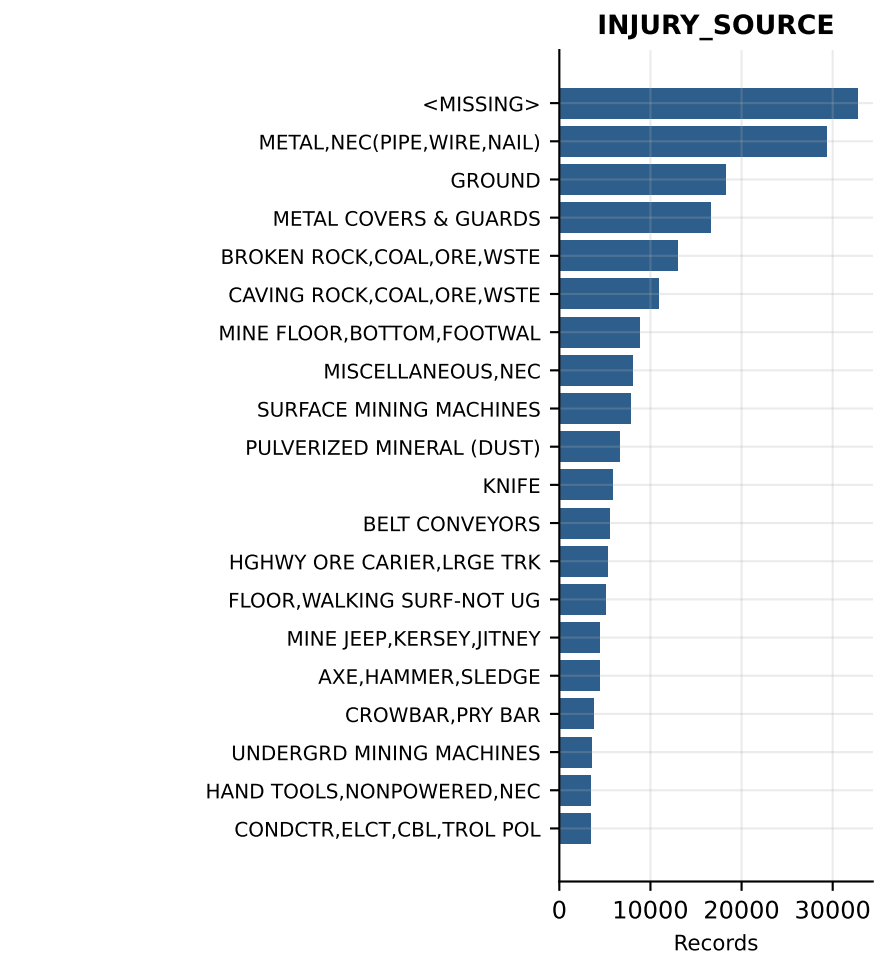


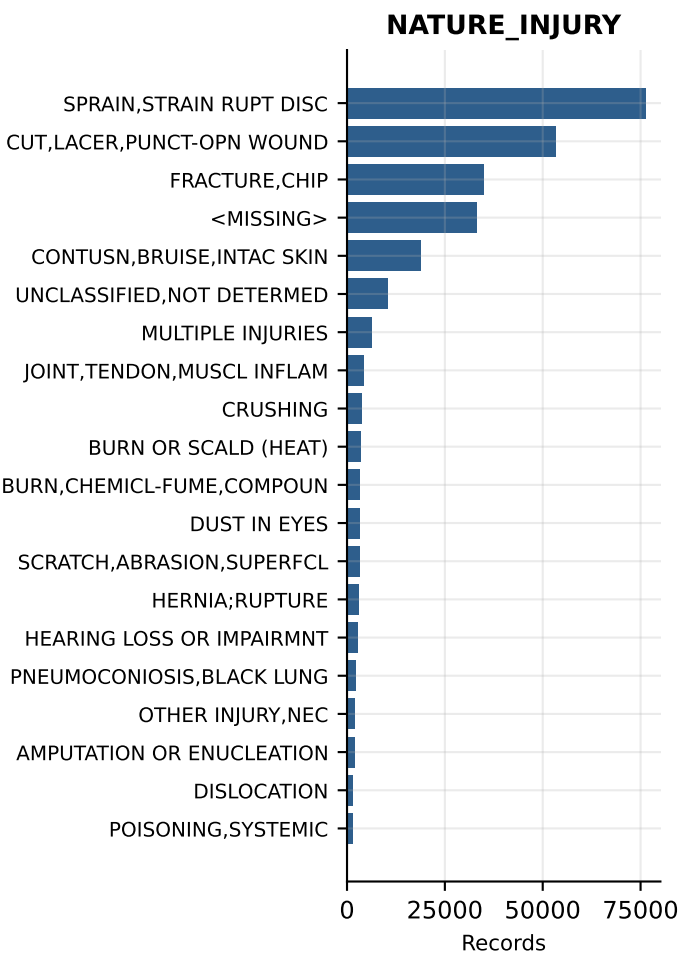


4. Categorical Variable Distributions

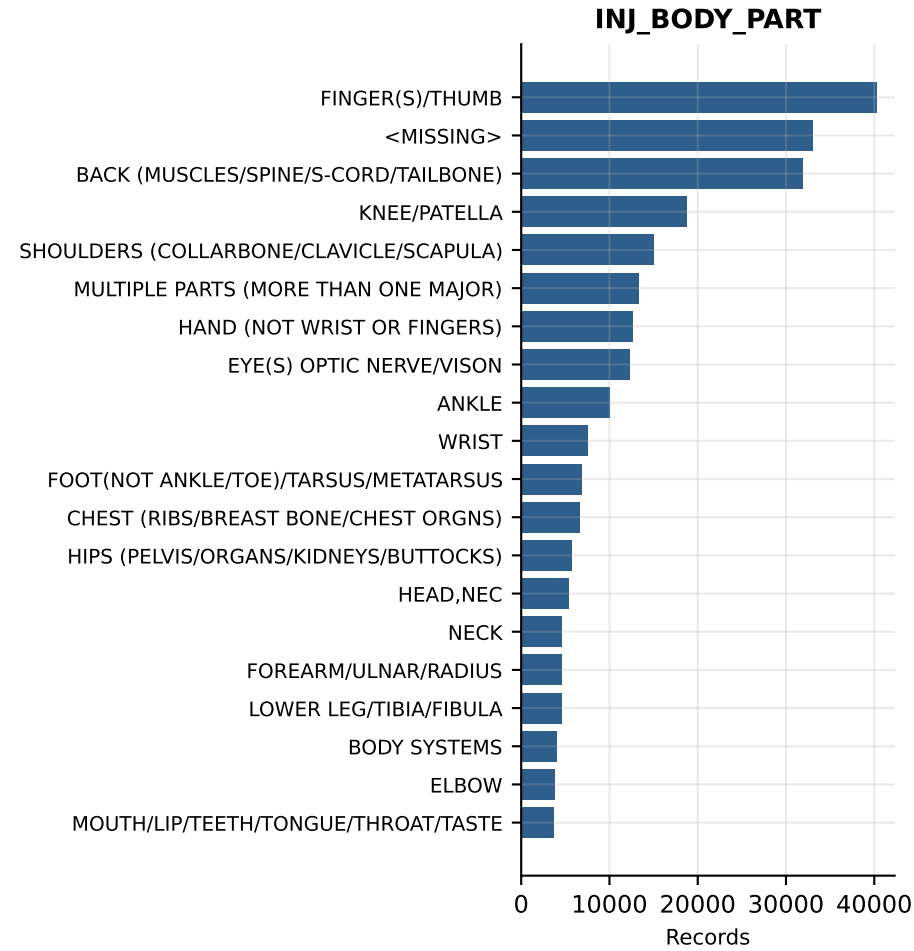


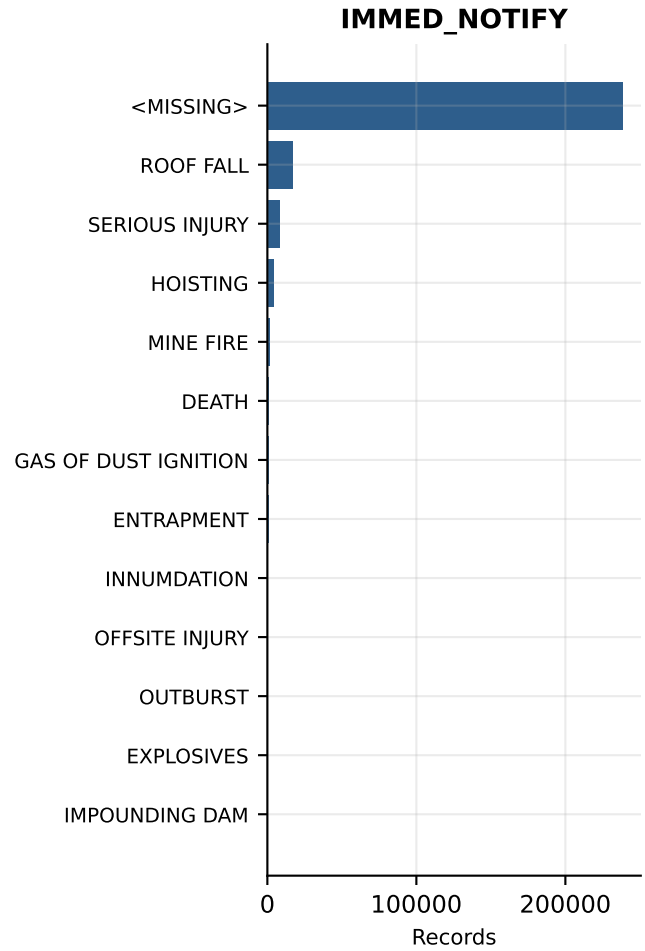


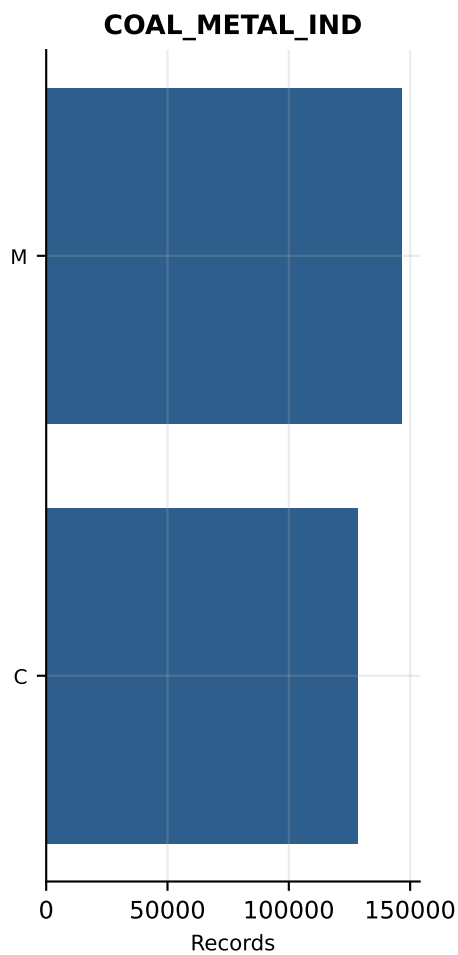


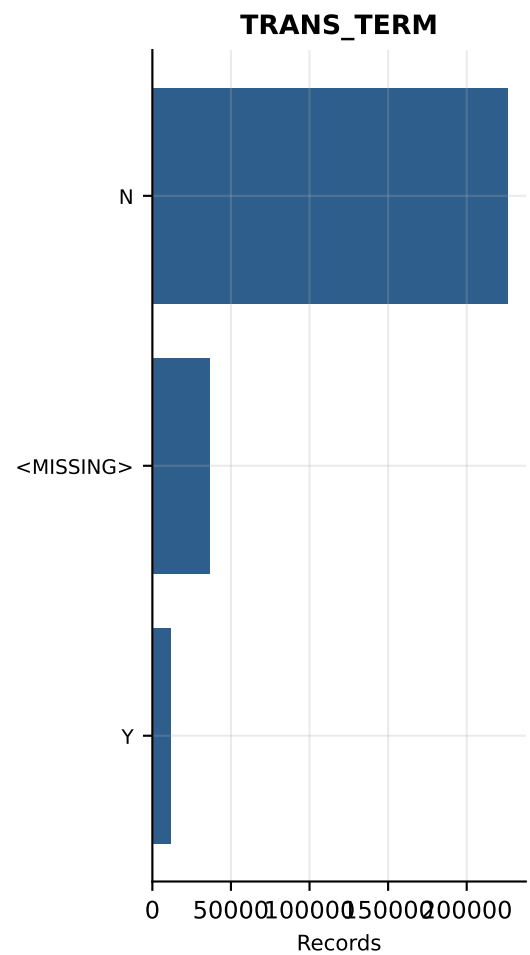


4. Categorical Variable Distributions





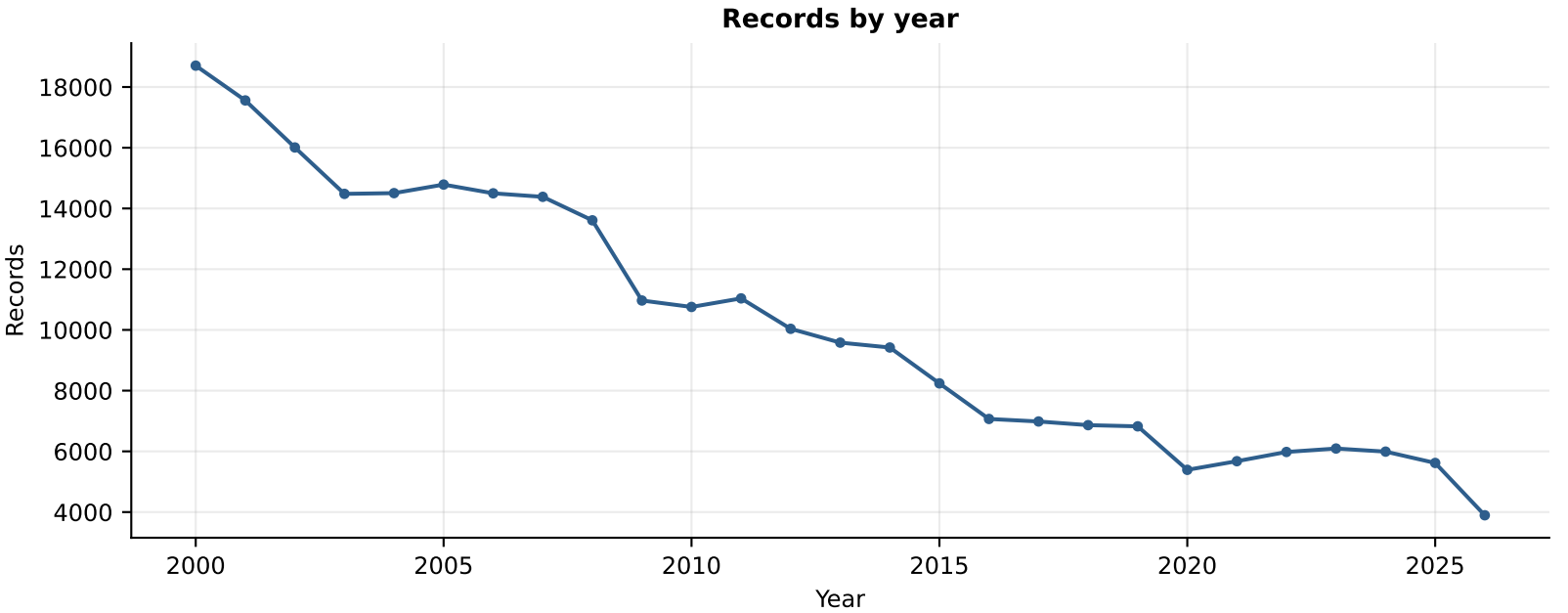


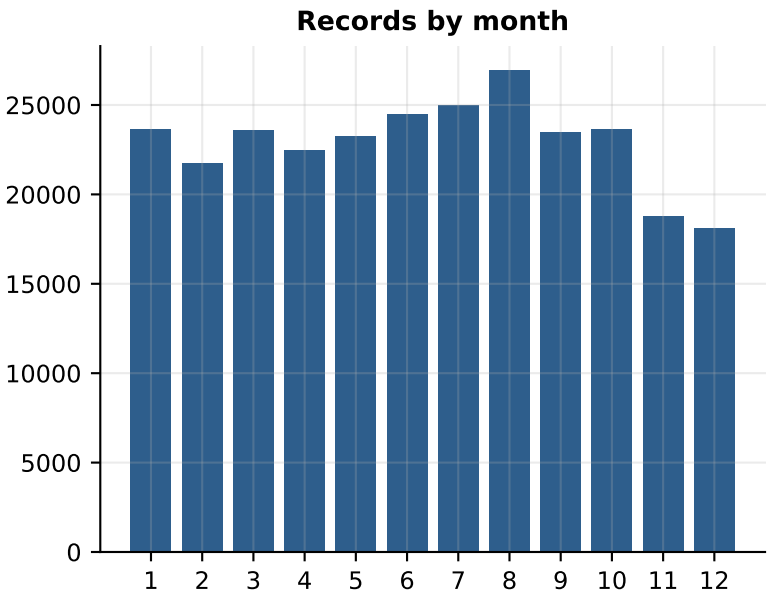


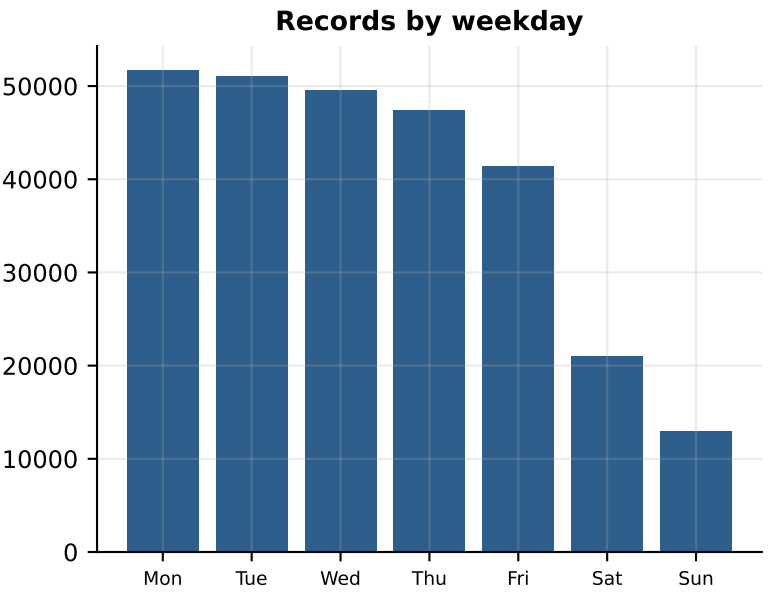
5. Code and Description Consistency

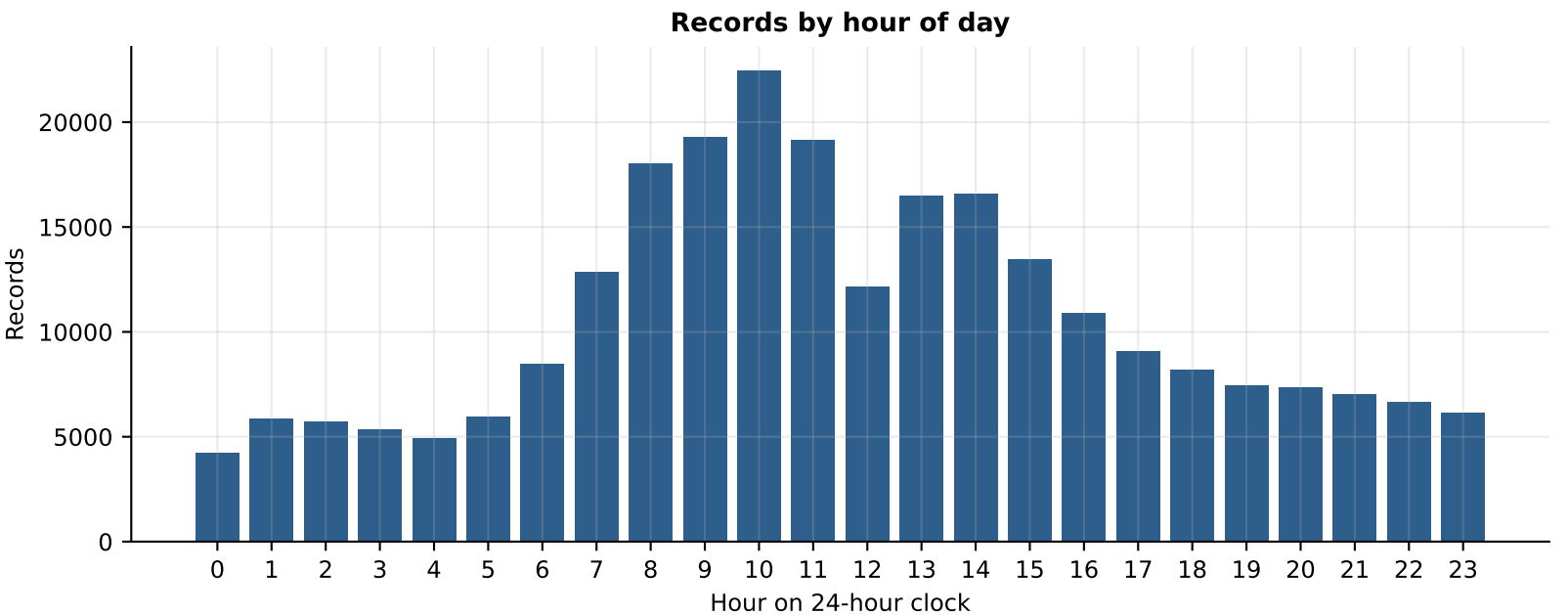
	code_column	label_column	unique_codes	unique_labels	codes_with_multiple_labels
0	SUBUNIT_CD	SUBUNIT	10	10	0
1	DEGREE_INJURY_CD	DEGREE_INJURY	11	11	0
2	UG_LOCATION_CD	UG_LOCATION	7	7	0
3	UG_MINING_METHOD_CD	UG_MINING_METHOD	7	7	0
4	MINING_EQUIP_CD	MINING_EQUIP	69	69	0
5	CLASSIFICATION_CD	CLASSIFICATION	28	28	0
6	ACCIDENT_TYPE_CD	ACCIDENT_TYPE	44	44	0
7	OCCUPATION_CD	OCCUPATION	255	203	0
8	ACTIVITY_CD	ACTIVITY	99	99	0
9	INJURY_SOURCE_CD	INJURY_SOURCE	127	127	0
10	NATURE_INJURY_CD	NATURE_INJURY	38	38	0
11	INJ_BODY_PART_CD	INJ_BODY_PART	46	46	0
12	IMMED_NOTIFY_CD	IMMED_NOTIFY	12	12	0

6. Temporal Accident Patterns









7a. Narrative Descriptive Statistics

```
NARRATIVE COMPLETENESS
--------------------------------------------------------------------------------
Narratives present ................. 274,943 of 274,955
Completion rate .................... 100.00%
Blank or missing ................... 12
Unique narratives .................. 273,332
Exact duplicate narratives ........ 1,611
Duplicate percentage ............... 0.59%

LENGTH IN CHARACTERS
--------------------------------------------------------------------------------
Mean / median ...................... 188.8 / 175.0
Minimum / maximum .................. 3 / 610
P90 / P95 / P99 .................... 331.0 / 363.0 / 382.0
At or near 384-character limit .... 2,312 (0.84%)

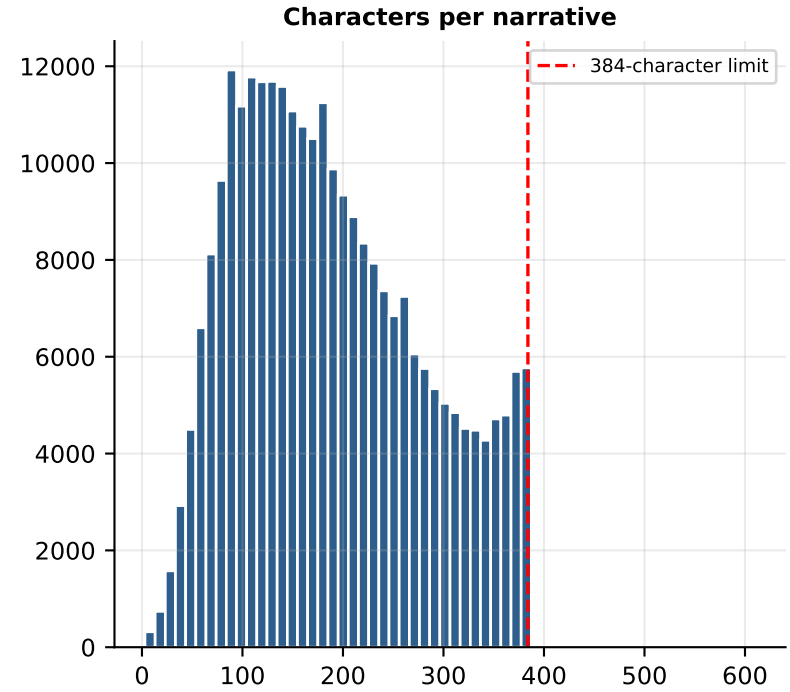
LENGTH IN WORDS
--------------------------------------------------------------------------------
Mean / median ...................... 34.1 / 32.0
Minimum / maximum .................. 1 / 115
Under five words ................... 1,676 (0.61%)

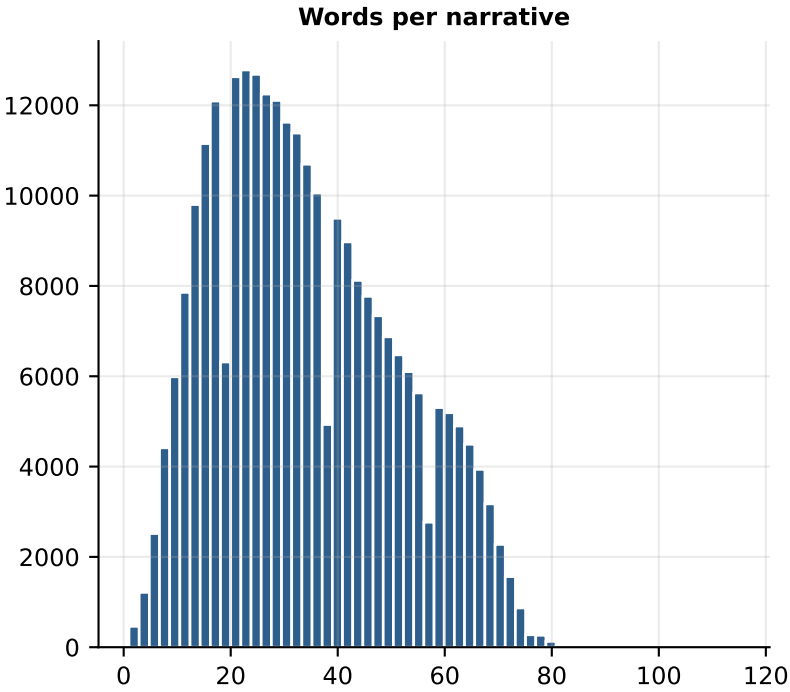
SENTENCE STRUCTURE
--------------------------------------------------------------------------------
Mean sentences per narrative ...... 2.41
Mean words per sentence ............ 15.97

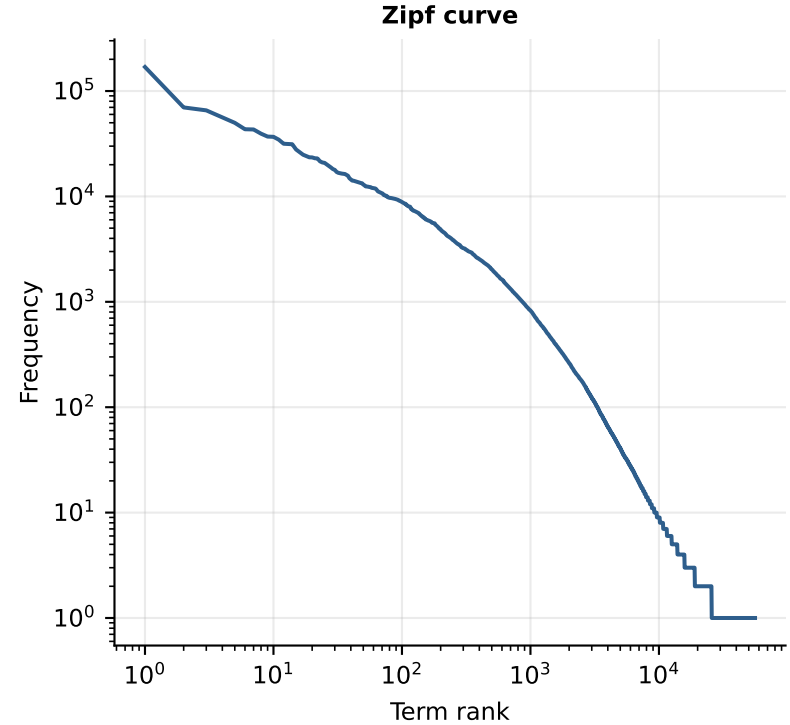
VOCABULARY
--------------------------------------------------------------------------------
Total usable tokens ................ 5,068,658
Unique terms ....................... 55,961
Type-token ratio ................... 0.01104
Terms occurring once ............... 30,341

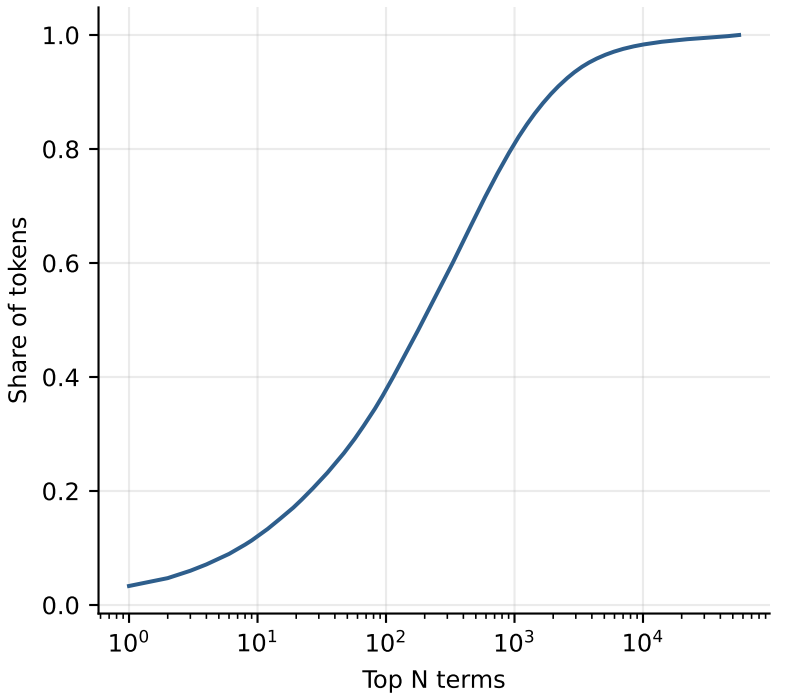
FORMATTING
--------------------------------------------------------------------------------
All-uppercase narratives ........... 81,634
Narratives containing digits ...... 125,955
Narratives containing non-ASCII ... 87
Narratives with repeated spaces ... 44,466
Without ending punctuation ........ 21,654
```

7b. Narrative Length and Vocabulary

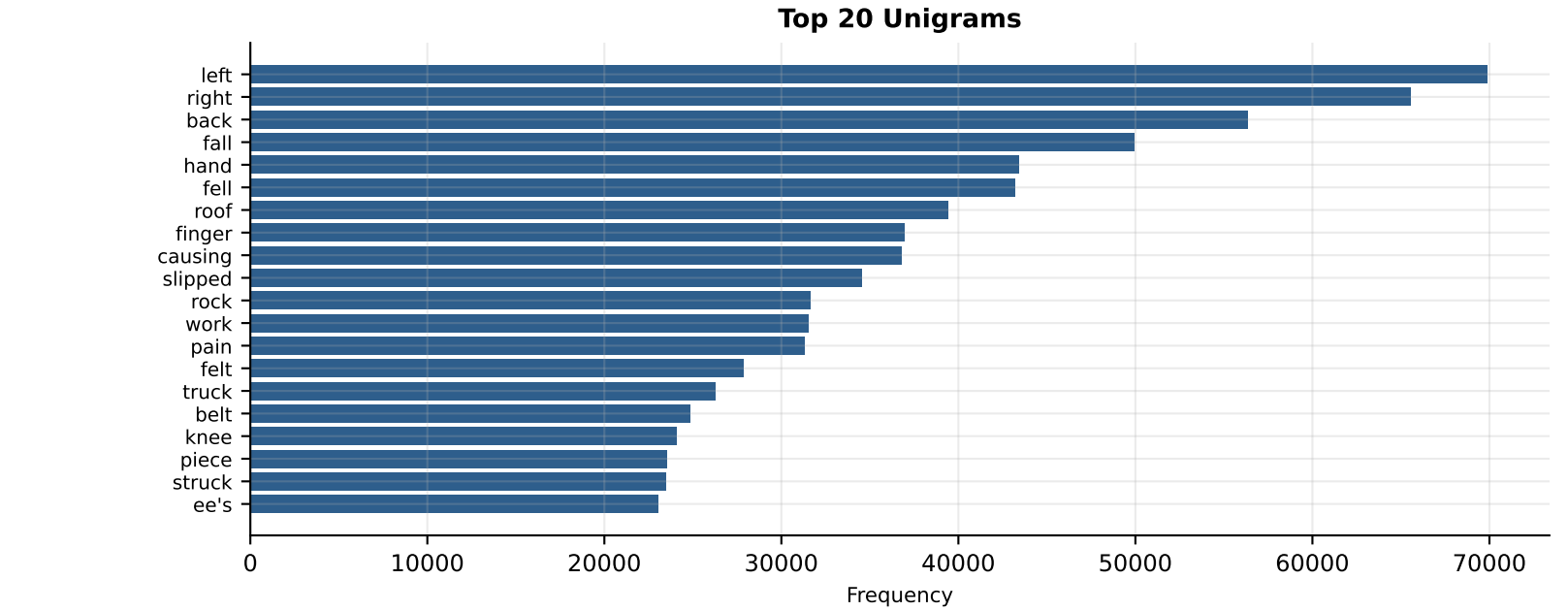


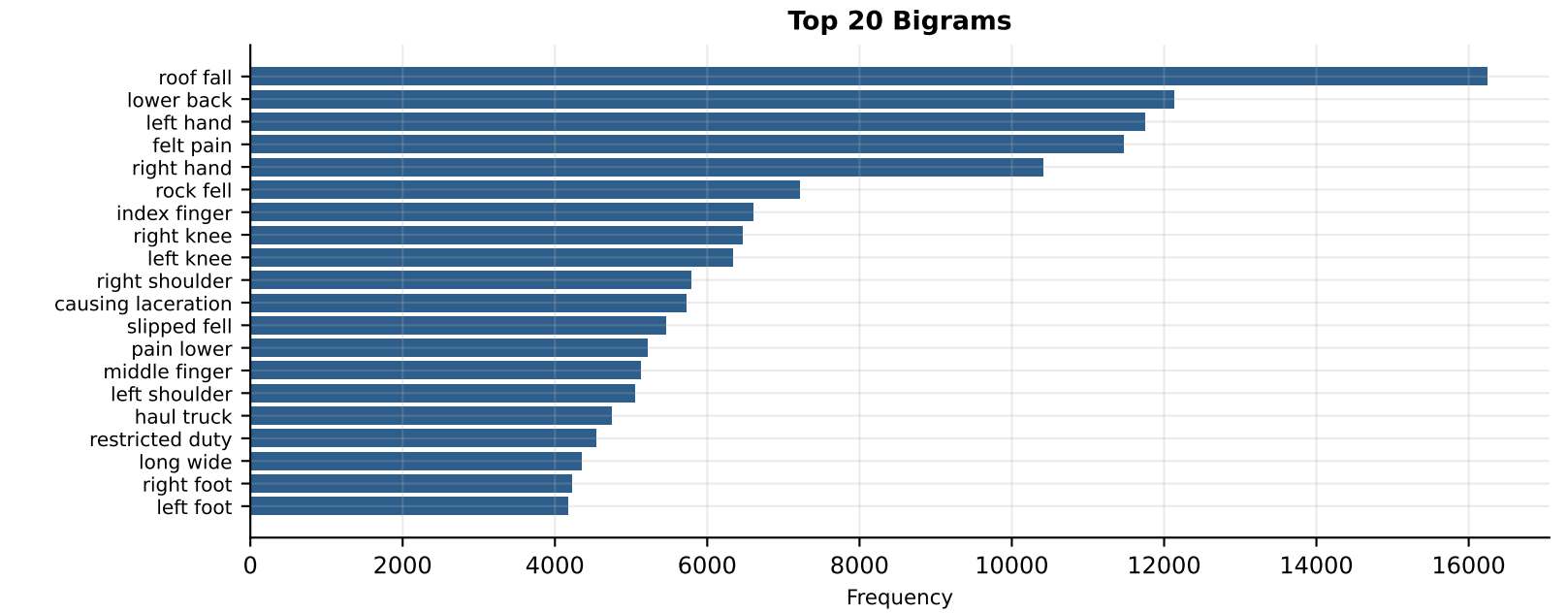


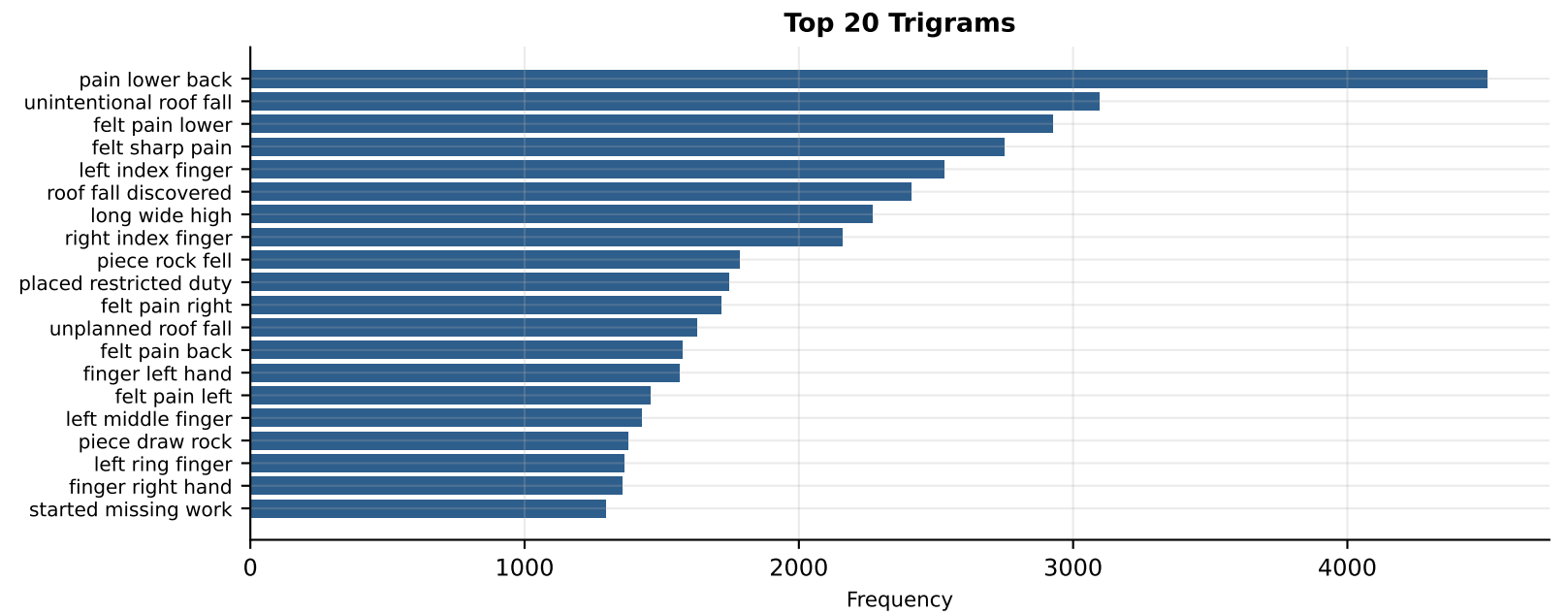




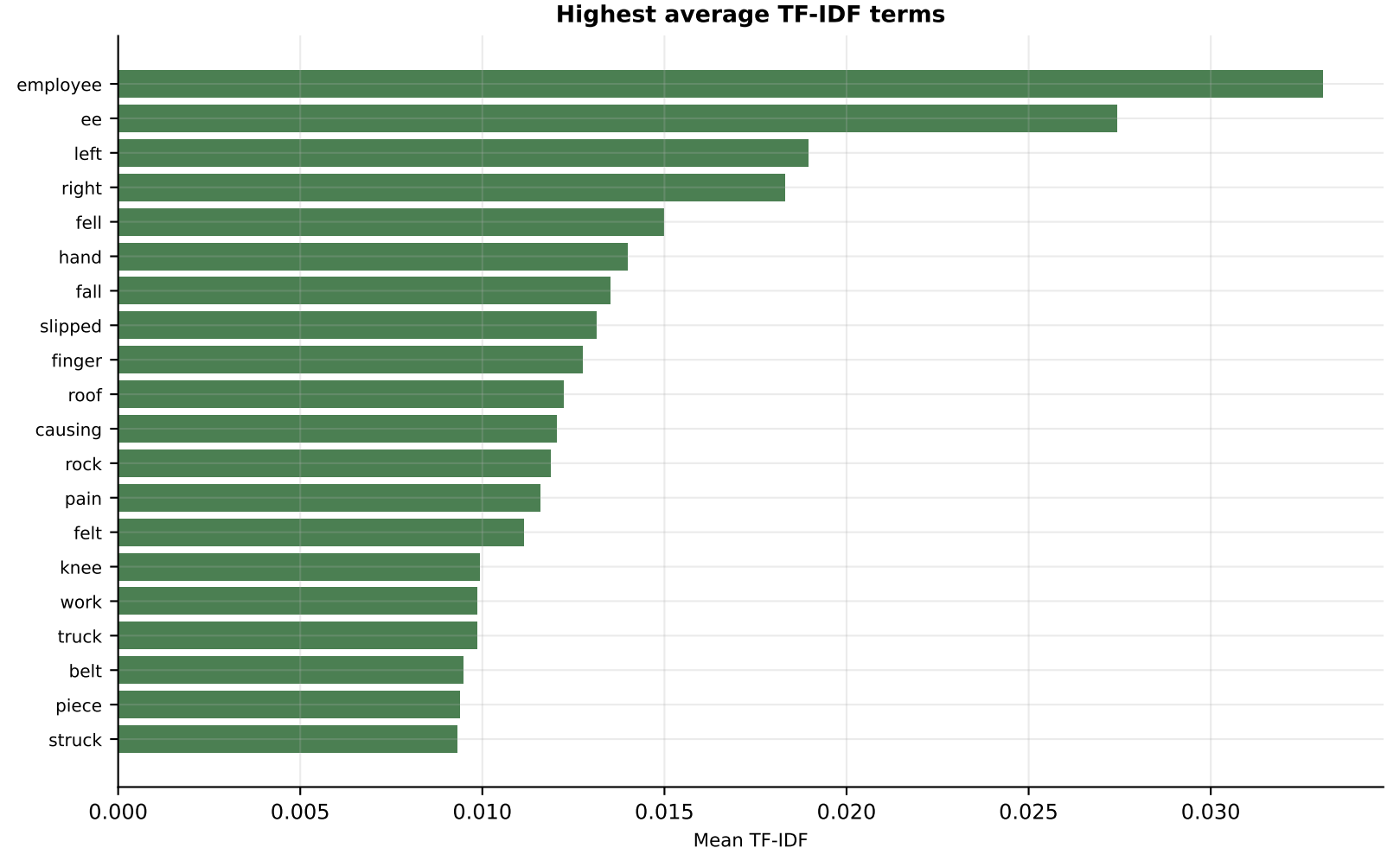
8a. Most Frequent Terms and Phrases







8c. Highest Mean TF-IDF Terms



8d. TF-IDF Matrix Characteristics

```
Document-term matrix shape ........ 274,943 x 20,000
Stored non-zero entries ........... 6,627,891
Matrix sparsity ................... 99.8795%
Mean non-zero features/document ... 24.1

Vectorizer settings
  Features: unigrams and bigrams
  English stopwords removed
  Minimum document frequency: 5
  Sublinear term-frequency scaling
  Maximum vocabulary size: 20,000
```

9a. NMF Topic Model: 8 Topics

TOPIC 1: 32,003 narratives (11.64%)
   finger, hand, employee, left, left hand, right, right hand, index, caught, index finger, middle, laceration

TOPIC 2: 23,381 narratives (8.50%)
   fall, roof, roof fall, entry, occurred, wide, fall occurred, long, 20, high, spad, area

TOPIC 3: 36,356 narratives (13.22%)
   pain, felt, felt pain, lower, employee, pain lower, shoulder, lifting, sharp pain, sharp, felt sharp, pain right

TOPIC 4: 50,098 narratives (18.22%)
   knee, employee, slipped, twisted, ankle, stepped, walking, foot, right, left, fell, right knee

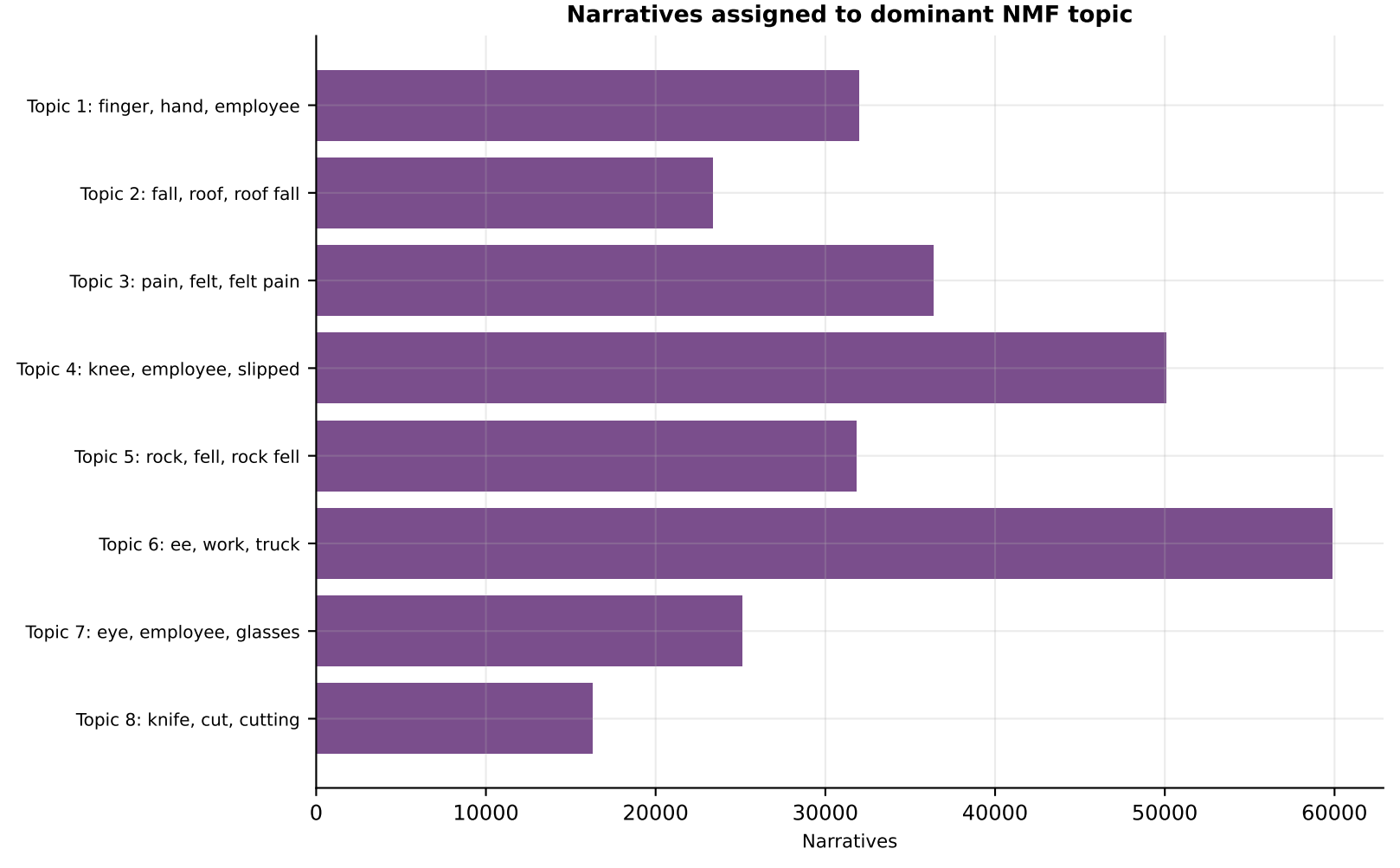
TOPIC 5: 31,850 narratives (11.58%)
   rock, fell, rock fell, piece, striking, roof, struck, piece rock, head, miner, bolt, bolter

TOPIC 6: 59,884 narratives (21.78%)
   ee, work, truck, duty, ee left, went, time, medical, ee right, did, hit, injury

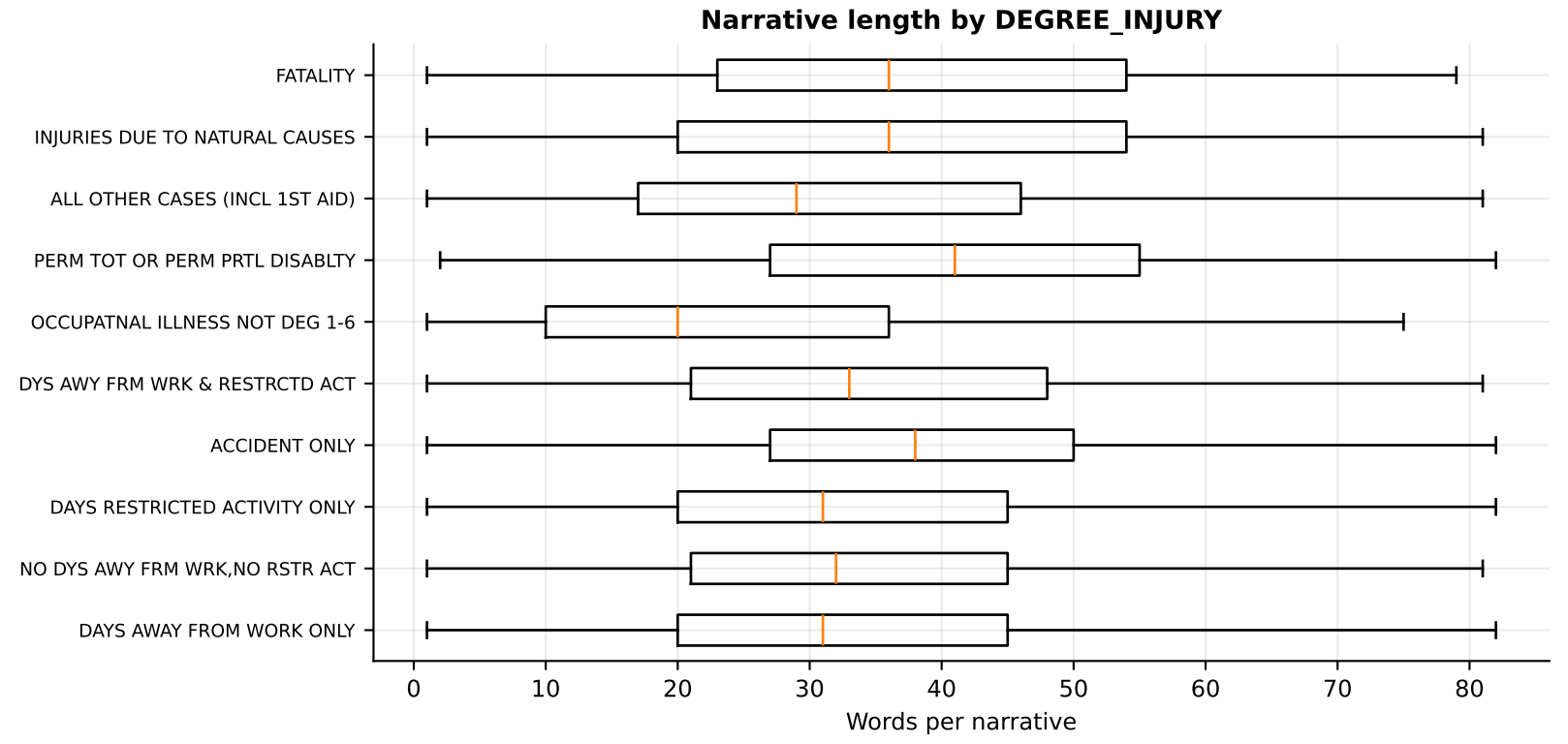
TOPIC 7: 25,104 narratives (9.13%)
   eye, employee, glasses, metal, safety, safety glasses, right eye, left eye, piece, dust, got, piece metal

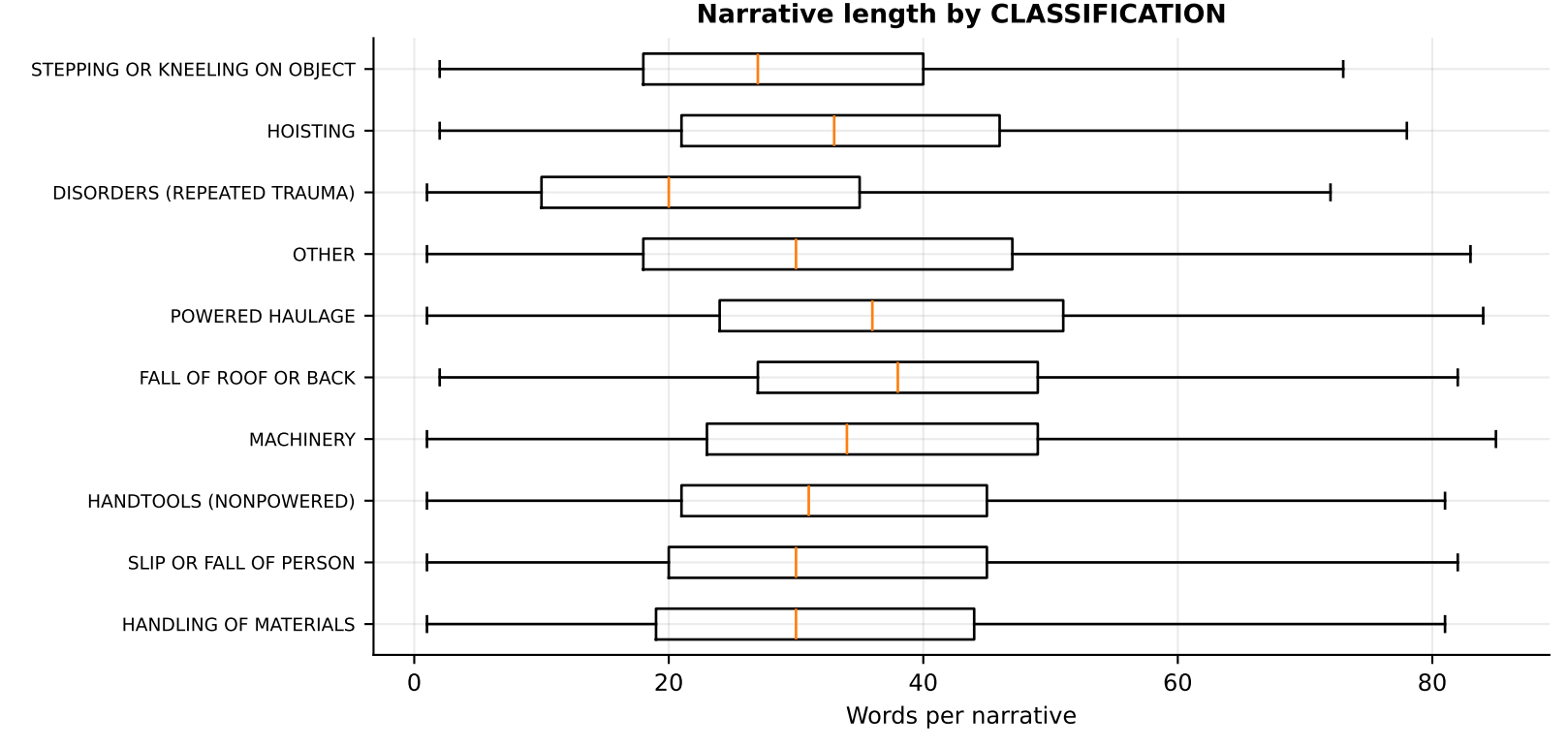
TOPIC 8: 16,267 narratives (5.92%)
   knife, cut, cutting, knife slipped, slipped, belt, utility, utility knife, slipped cut, stitches, cut left, employee cutting

9b. Topic Prevalence



10a. Narrative Length by Incident Outcome





10b. Random Sample of Narratives

[Sample 1]
Water truck driver reported a noise while steering to the right. Pit mechanic installed new grease lines, the mechanic instructed the water truck driver to steer the wheels from side to side as grease was added. As the wheels were turned to the right the mechanic was pinned between the tire and frame causing injuries to the mechanic, 4 broken ribs and a broken collar bone.

[Sample 2]
Roof Fall #4 entry spad # 860. Fall was about 20' X 30' X 5' thick. Fell up to the coal rider.

[Sample 3]
Employee was assisting in opening the tarp on a truck to load by standing on the fuel tank. The tank was covered in ice causing the employee to slip and fall to the ground landing on both feet but spraining the right knee upon landing.

[Sample 4]
Employee went to remove lock out and turn breaker on. When he turned on the rod mill breaker it blew up blowing breaker doors open. He burnt his right arm and bruised right forefinger. He was taken to co dr and prescribed an antibiotic ointment for the burn.

[Sample 5]
Miner was raking ash at kiln hood when a piece of ash hit the end of the rake causing the rake to hit the miner on the chin. Miner sustained a laceration requiring stitches.

[Sample 6]
EE WAS CUTTING BACK BRUSH WITH WEED WACKER, CONTRACTED POISON IVY.

[Sample 7]
AN IGNITION ON THE FACE OF THE LONGWALL PANEL IN 13 BUTT WEST #44 PANEL OCCURRED AT APPROX. 1:05 AM. THE IGNITION HAPPENED WHILE MINING WAS IN PROGRESS FROM THE HEADGATE TO THE TAILGATE AT 126 SHIELD. A JOY 7LS SHEARER WAS BEING USED. THE FLAME LASTED BETWEEN EIGHT AND TEN SECONDS AND WAS EXTINGUISHED BY WATER FROM THE TAIL DRUM SPRAYS. NO INJURIES.

[Sample 8]
The ee complained of having a sore left knee and sore elbows from stacking 50 lb bags of sand.

11. Findings and Modeling Recommendations

```
DATA-QUALITY FINDINGS
--------------------------------------------------------------------------------
Columns over 50% missing ........... 13
  CONTRACTOR_ID, IMMED_NOTIFY, EQUIP_MODEL_NO, INVEST_BEGIN_DT, UG_MINING_METHOD_CD, UG_MINING_METHOD, UG_LOCATION, UG_LOCATION_CD, CLOSED_DOC_NO, EQUIP_MFR_CD, EQUIP_MFR_NAME, MINING_EQUIP, MINING_EQUIP_CD
Narratives near 384-char limit .... 2,312
Exact duplicate narratives ........ 1,611

DATA-HANDLING CONSIDERATIONS
--------------------------------------------------------------------------------
* Keep identifiers as strings so leading zeros are preserved.
* Review code and description pairs before using both in one model.
* Validate whether ACCIDENT_TIME = 0 means midnight or not recorded.
* Keep duplicate narratives within the same train/test partition.
* Treat narratives near the source-field limit as potentially truncated.

MODELING RECOMMENDATIONS
--------------------------------------------------------------------------------
* Start with TF-IDF plus logistic regression or a linear SVM.
* Candidate targets include DEGREE_INJURY, CLASSIFICATION, ACCIDENT_TYPE,
  INJURY_SOURCE, and INJ_BODY_PART.
* Use stratified evaluation and report macro-F1, precision, and recall.
* Compare structured-only, narrative-only, and combined models.
* Consider grouping train/test splits by MINE_ID to test generalization.
* Treat NMF topics as exploratory themes requiring subject-matter review.
```